2021 WSMO Team Round

SMO Team

TR 1: How many strings can be formed by selecting, without replacement, three letters from the phrase "Winter Solstice" (ignore the space) and arranging them, such that the first letter is an uppercase letter and the next two letters are lowercase?

TR 2: Bobby has some pencils. When he tries to split them into 5 equal groups, he has 2 left over. When he tries to split them into groups of 8, he has 6 left over. What is the second smallest number of pencils that Bobby could have?

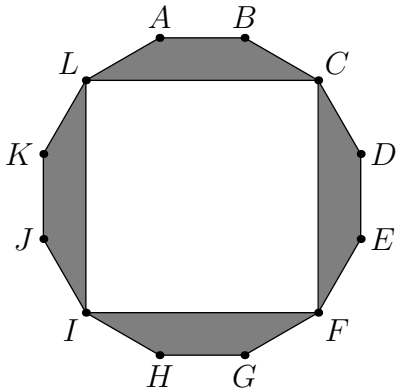
TR 3: Farmer Sam has n dollars. He knows that this is exactly enough to buy either 50 pounds of grass and 32 ounces of hay or 96 ounces of grass and 24 pounds of hay. However, he must save 4 dollars for tax. After some quick calculations, he finds that he has exactly enough to buy 18 pounds of grass and 16 pounds of hay (and still have money left over for tax!). Find n.

TR 4: Consider a triangle $A_1B_1C_1$ satisfying $A_1B_1 = 3, B_1C_1 = 3\sqrt{3}, A_1C_1 = 6$. For all successive triangles $A_nB_nC_n$, we have $A_nB_nC_n \sim A_{n-1}B_{n-1}C_{n-1}$ and $A_n = B_{n-1}, C_n = C_{n-1}$, where $A_nB_nC_n$ is outside of $A_{n-1}B_{n-1}C_{n-1}$. Find the value of

$$\left(\sum_{i=1}^{\infty}[A_iB_iC_i]\right)^2.$$

TR 5: Two runners are running at different speeds. The first runner runs at a constant 12 miles per hour. The second runner runs at $t + 4$ miles per hour, where t is the number of hours that have passed. After $n > 0$ hours, the runners have run the same distance. Find n.

TR 6: Suppose that regular dodecagon $ABCDEFGHIJKL$ has side length 5. The area of the shaded region can be expressed as $a + b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a + b + c$.



TR 7: A frog makes one hop every minute on the first quadrant of the coordinate plane (this means that the frog's x and y coordinates are positive). The frog can hop up one unit, right one unit, left one unit, down one unit, or it can stay in place, and will always choose a valid hop from these 5 directions (a valid hop is a hop that does not place the frog outside the first quadrant) uniformly at random. Given that the frog starts at $(1, 1)$, the expected number of minutes until the frog reaches the line $x + y = 5$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

TR 8: Isaac, Gottfried, Carl, Euclid, Albert, Srinivasa, Rene, Adihaya, Euler, and Hypatia sit around a round table. There are $a \cdot b!$ possible seatings (disregarding rotations) where Euler isn't seated next to Hypatia and Isaac isn't seated next to Gottfried, where b is maximized and both a and b are positive integers. Find $a + b$.

TR 9: In triangle ABC, points D and E trisect side BC such that D is closer to C than E. If $\angle CAD = \angle EAD, ED = 3$, and $[AEB] = 6$, then find $[ABC]$.

TR 10: The minimum possible value of

$$\sqrt{x^2 + y^2} + \sqrt{3x^2 + 3y^2 - 6x + 12y + 15}$$

can be expressed as $\sqrt{m}$. Find m.

TR 11: Find the remainder when

$$\sum_{\substack{x+y+z \leq 10 \\ x,y,z \in \mathbb{N}_0}} \frac{(x + y + z)!}{x!y!z!}$$

is divided by 100.

TR 12: Let $x, y, z \in \{0, 1, \ldots, 99\}$ be chosen independently and uniformly at random. The probability that the sum of the factors of $2^x 3^y 5^z$ is divisible by 6 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

TR 13: Square $BCDE$ is drawn outside of equilateral triangle ABC. Regular hexagon $DEFGHI$ is drawn outside of square $BCDE$. If the area of triangle AED is 3, then the area of triangle AGH can be expressed as $a\sqrt{b} - c$, where a, b, and c are positive integers and b is not divisible by the square of any prime. Find $a + b + c$.

TR 14: Suppose that x is a complex number such that $x + \frac{1}{x} = \frac{\sqrt{6}+\sqrt{2}}{2}$ and the imaginary part of x is nonnegative. Find the sum of the five smallest nonnegative integers n such that $x^n + \frac{1}{x^n}$ is an integer.

TR 15: Let $ABCD$ and $DEFG$ be squares that intersect exactly once and with areas 1011^2 and 69^2 respectively. Find the largest real number M such that $CE + AG > M$ for all such pairs of squares.